



ExAq-MSPP: An Energy-Efficient Mobile Sink Path Planning Using Extended Aquila Optimization Algorithm

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Abstract

Wireless sensor networks play a crucial role in gathering data from remote or hard-to-reach locations, enabling real-time monitoring and decision-making in a wide range of industries and applications. The mobile sink path planning (MSPP) enables mobile sinks (e.g., drones or rovers) to navigate through the environment, collecting data from different sensor nodes, ensuring comprehensive coverage, and adaptively addressing changing conditions. Still, the energy-efficient routing with minimal delay is the challenging aspect. This research focuses on improving data gathering in wireless sensor networks by introducing an efficient routing protocol. In this proposed protocol, sensor nodes are initially deployed using Voronoi diagrams to ensure uniform network coverage. The network is then divided into clusters using the low-energy adaptive clustering hierarchy (LEACH) algorithm for energy-efficient routing. To optimize the path planning of a mobile sink for data collection, we introduce the extended Aquila (ExAq) optimization algorithm, which uses a multi-objective fitness function considering factors such as delay, residual energy, link quality, priority, and distance. Simulation results demonstrate the effectiveness of the proposed ExAq-MSPP protocol in terms of reduced delay, improved network lifetime, higher packet delivery ratio, enhanced residual energy, and increased throughput compared to existing protocols with the values of 1.169, 99.857, 99.920, 0.997, and 255.306, respectively. Thus, the energy-efficient routing and optimizing path planning for mobile sinks, the proposed ExAq-MSPP protocol can extend network lifetime, increase data accuracy, and provide more robust performance under changing environmental conditions.

Keywords Wireless sensor networks · Extended Aquila-based routing · Mobile sink path planning · Energy-efficient LEACH-based clustering · Network lifetime

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1 Introduction

Wireless sensor networks (WSNs) represent a revolutionary technology, connecting a web of sensors with wireless communication capabilities across various sectors, including environmental monitoring, industrial automation, and healthcare [1–4]. Within the realm of WSNs, one critical challenge that stands out is the coverage issue. Coverage pertains to the effectiveness of monitoring within the sensor-deployed area [5]. The presence of coverage gaps or overlaps can result in inefficiencies and missed data events. To mitigate these concerns, Voronoi-based node deployment emerges as a key technique [6]. In Voronoi-based node deployment, strategic placement of sensor nodes is employed to optimize coverage. The foundation of this concept lies in Voronoi diagrams, which meticulously partition the sensing area into distinct regions, each assigned to a specific sensor node [7]. The benefit of this

approach is that it ensures that every point within the sensing area is under the watchful eye of at least one sensor node [8, 9]. By adopting this methodology, we can effectively minimize coverage issues such as gaps and overlaps. Thus, a more efficient and reliable data collection process within the WSN [10, 11].

Small sensor nodes make up WSNs, which are typically battery-powered and tasked with collecting data from their allocated settings [12]. These nodes exchange information wirelessly and send it to a centralized sink node [13]. However, energy consumption is a serious problem, particularly in vast networks with a large number of sensor nodes [14, 15]. The key factor is battery life because charging is frequently impracticable, especially in dangerous or difficult-to-reach areas. Energy usage and network durability are now top research priorities as a result of this scenario [16]. The clustering-based technique is a clever way to save energy on sensor nodes and prolong the life of the network [17]. According to this plan, the network is split into clusters, each of which is headed by a cluster head (CH) [18]. These cluster heads are crucial because they gather and aggregate data from sensor nodes under their jurisdiction before sending it directly or indirectly through other cluster heads to the central sink node [19]. The researchers introduced a cluster-based routing method utilizing mobile sinks to enhance energy efficiency in wireless sensor networks. By deploying mobile sinks, the approach balances energy consumption across nodes, extending network lifespan and improving reliability [20]. The researchers proposed a two-level clustering method for IoT networks, combining fuzzy logic and content-based routing. The approach optimizes data transmission by clustering devices based on content relevance, improving communication efficiency and reducing network traffic [21].

The use of sink nodes is one of the most popular ways to improve communication quality inside networks. Sink nodes, whether mobile or static, play a crucial role [22]. The problem with static sink nodes is that they remain fixed in place, which requires surrounding nodes to use more energy owing to the continuous data flow. Mobile sinks provide a solution by effectively traversing the network and gathering data from each node [23, 24]. This investigation purposefully uses a mobile sink node for data gathering to make use of mobile sinks' benefits [25]. Mobile sinks greatly lower the overall network's energy usage, but they also add some delay [4, 26]. Since the sink node is movable, data travel farther to it. The designed method incorporates the idea of clustering to address this latency concern [27]. In a single cluster that is headed by a cluster head, all nodes are gathered. The cluster head, in turn, effectively connects with the mobile sink thanks to this setup, allowing for continuous communication between participating nodes and both [28].

1.1 Motivation

The motivation for this research article stems from the critical role that WSNs play in collecting data from remote or difficult-to-access locations, which is essential for real-time monitoring and decision-making across various industries. Despite their potential, achieving energy-efficient routing with minimal delay remains a significant challenge. This research aims to address these issues by introducing an efficient routing protocol that leverages Voronoi diagrams for uniform network coverage and the low-energy adaptive clustering hierarchy (LEACH) algorithm for energy-efficient routing. In addition, the study proposes the extended Aquila (ExAq) optimization algorithm to optimize the path planning of mobile sinks, ensuring comprehensive coverage and adaptability to changing conditions. The motivation is to enhance the effectiveness of WSNs by reducing delays, extending network lifetimes, and improving data accuracy, thereby contributing to more reliable and efficient real-time monitoring systems. The major contribution of the research is:

- *Design extended Aquila (ExAq) optimization algorithm:* the proposed ExAq algorithm is designed by incorporating the Chebyshev mapping with the conventional Aquila optimization algorithm for obtaining the global best solution by eliminating the local solution trapping.
- *Design of ExAq-MSPP:* the mobile sink path planning (MSPP) is employed optimally using the ExAq algorithm by considering the multi-objective fitness function based on delay, residual energy, link quality, priority, and distance.

The organization of the research is as follows: Sect. 2 explains the literature review along with the problem statement. Section 3 details the ExAq-MSPP approach and its simulation outcome is detailed in Sect. 4. Finally, Sect. 5 concludes the research.

2 Literature Review

The review of conventional MSPP-based techniques of WSN is detailed in this section. A hybrid optimization-based MSPP was designed by [1, 2] for the heterogeneous WSN. In this, clustering-based routing was devised through the distribution density factor of neighboring nodes and average residual energy factor optimization of the stable election protocol (SEP) algorithm. Then, the MSPP was employed using the hybrid optimization algorithm that was capable of determining the best path-planning strategy for

the mobile sink, taking into account the optimization of edges and nodes in the minimum spanning tree. The result was an optimal path for the sink, ensuring efficient data collection in the heterogeneous WSNs. The SEP algorithm must balance energy efficiency with the need for certain QoS parameters, such as low latency or high reliability. Striking this balance is challenging, as optimizing for one aspect may negatively impact another.

Area priority-based MSPP was designed by Bagais et al. [29] to deliver urgent messages to the mobile sink without latency and with minimal packet loss. In this, an optimization approach was utilized for the clustering of the nodes. Then, the MSPP was employed to move towards zones with higher priority and shorter distances, guided by the optimization approach. Finally, for urgent messages originating from other zones that the mobile sink (MS) had traversed, the proposed approach employed a routing technique utilizing multi-hop communication based on the MS's current position, with PSO aiding in the routing decisions. Simulation results demonstrated that the introduced method outperformed other models based on different parameters, particularly in scenarios involving controlled MS movement in large-scale environments with a focus on handling urgent messages.

The introduced method involved the use of multiple portable sinks for efficient information gathering and integrated energy-balanced clustering with an artificial bee colony-based data-gathering strategy [30]. To determine the most suitable node to serve as the center of gravity for each cluster, the remaining energy of each sensor node was considered. This approach ensured that clusters were formed in an energy-efficient manner, enhancing the network's overall performance. Then, the MSPP, using the optimization algorithm, was capable of reducing data communication losses, extending the network's lifetime, conserving energy, maintaining system reliability, and enhancing network efficiency. The method assumes a certain tolerance for data delay, which may not be suitable for all applications. Some real-time or critical applications may require lower latency.

An optimization-based clustering and MSPP routing was designed by Umamaheswari & Kumar [31] to address the selection of CHs and determine optimal paths for mobile sinks. In the first step, the technique utilized a method based on the optimization algorithm for clustering. This process aimed to efficiently organize sensor nodes into clusters, with CHs strategically chosen within each cluster. The selection of CHs was based on energy-aware criteria and other relevant parameters to optimize the clustering process. The second major component of the technique involved weighted sum approach (WSA)-based path planning for MSs. This path-planning approach was designed to enable mobile sinks to navigate from their current location to their destination in an optimal manner.

The WSA technique considered various factors to calculate the most efficient paths for the MSs, taking into account energy constraints and other parameters. Overall, the introduced model aimed to enhance the performance of WSNs by intelligently selecting CHs and guiding MSs along optimized paths. This approach contributed to improved energy efficiency, reduced data transmission delays, and enhanced overall network performance.

The framework introduced by El-Fouly et al. [32] utilized a path-planning technique that takes into account several crucial factors, including the priority of different areas, environmental parameters, and the urgency of messages. As a part of this framework, a novel clustering algorithm was presented. This algorithm focused on optimizing energy efficiency and reliability within the network. It achieved this by considering critical factors such as the remaining energy levels of sensor nodes, the quality of wireless links, and the average intra-cluster distance. In addition, the research proposed a real-time, energy-efficient, and reliable routing algorithm that was also environmentally aware. This routing algorithm factored in environmental data, link quality, communication delay, hop count, the residual energy of nodes, and load balancing to make informed routing decisions. The results unequivocally demonstrated that the proposed framework consistently outperformed the existing algorithms. Incorporating diverse factors into a cohesive and efficient path-planning strategy poses a significant challenge.

An improved sparrow search algorithm (MISSA) with the dynamic window approach (DWA) was designed by Hou et al. [33] for dynamic path planning. The designed algorithm utilized logistic-tent chaotic mapping for a well-distributed initial population. In addition, the escaping strategy of the model was further enhanced through the refined evaluation function for improved safety and adaptive velocity adjustment strategy. The designed model achieved smooth and safe path generation with dynamic obstacle avoidance in MISSA-DWA. The consideration of distance as the fitness parameter was the challenging aspect.

Rendezvous points (RPs) and fuzzy logic (Fuzzy_RP)-based adaptive path construction to enhance data-gathering efficiency and prolong network lifetime by optimizing the path of a mobile sink was designed by Banimelhem et al. [34]. Initially, a set of RPs was selected using the k-means algorithm. Then, Fuzzy_RPs was employed with three inputs (remaining energy, transmission distance, number of neighbors) and one output (weight value). Calculation of weight values for each sensor node based on Fuzzy inputs and updated RP locations based on these weight values. The mobile sink moves sequentially between the updated RPs to collect data. The scalability issue of the model was considered as a challenging aspect. The short description is presented in Table 1.

Table 1 Short description

References	Technique used	Advantages	Disadvantages
[1, 2]	Pigeon-inspired optimization algorithm	Low latency and high reliability	Demand for significant computational resources and potentially results in increased processing time
[30]	Energy-efficient path planning of MS-based area priority	The prioritization approach enhances the system's adaptability to different environmental conditions and varying levels of importance for different zones	Failure to handle significant parameters that minimize the latency of the model
[30]	Artificial bee colony	The optimization algorithm is designed to reduce data communication losses, leading to an extension of the network's lifetime	Real-time or critical applications that require lower latency may not be well-supported by this approach, posing a limitation in its applicability
[31]	Energy-aware metaheuristics-based path planning with mobile sinks	Energy-efficient clustering and routing contribute to reduced energy consumption, making the network more sustainable and resilient	As the size of the network grows, scalability concerns may arise, impacting the model's effectiveness in large-scale deployments
[32]	The energy-efficient and reliable clustering algorithm	Well-balanced clusters that enhance network performance and sustainability	Consideration of multiple parameters poses challenges in ensuring timely decision-making while maintaining energy efficiency and reliability
[33]	Dynamic path planning using a hybrid metaheuristic algorithm	Robust path optimization capability	Limited to considering only distance in path planning
Banimelhem et al. [34]	Adaptive path planning using fuzzy logic	Improved data-gathering efficiency by adapting RPs to network conditions	Potential communication overhead for updating RP locations
Proposed	ExAq-MSPP	(i) Achieved a lower delay of 1.169 s, enhancing the real-time data-gathering capability (ii) Achieved a packet delivery ratio of 99.920%, indicating reliable data transmission (iii) Maintained a higher residual energy of 0.997 Joules, ensuring longer operational periods for sensor nodes	–

2.1 Problem Statement

Mobile sink path planning is a critical aspect of wireless sensor networks that involves determining the most efficient route or trajectory for a mobile sink device to traverse within the network while collecting data from sensor nodes. This process is fundamental to optimizing data collection in WSNs and plays a pivotal role in various application domains. Application domains for mobile sink path planning encompass environmental monitoring, precision agriculture, healthcare, industrial automation, and disaster response, among others. In these domains, mobile sinks are employed to efficiently collect data from sensors scattered across vast or dynamic environments.

Existing methods for optimal mobile sink path planning span various approaches, including greedy algorithms, heuristic algorithms, and optimization techniques such as linear and integer programming. However, despite the progress in this field, significant challenges persist. These challenges include adapting to dynamic environments where path planning strategies must respond to changing conditions, ensuring energy-efficient data collection to extend the network's lifetime, meeting real-time data-collection requirements, scaling path planning methods for large sensor networks, ensuring reliable data collection in the presence of communication failures, balancing the data collection load across sensor nodes, and accommodating path constraints such as physical obstacles or restricted areas. Addressing these challenges is crucial for advancing the efficiency and effectiveness of mobile sink path planning in WSNs. Thus, an energy-efficient clustering and optimal path planning by

considering the multi-objective fitness function is required for overcoming the challenges faced by the conventional methods.

3 System Model

The WSN comprises various sensor nodes, wherein each node is distributed randomly. Voronoi diagrams partition the network area into regions, with each sensor node being the focal point of its region. This ensures that the deployment provides uniform coverage of the entire area, reducing the chances of coverage gaps or redundant nodes in certain areas. The sink gathers the data generated by the nodes through the CH. Since the CH-based data collection minimizes the communication link between the nodes and the sink, network congestion can be minimized through CH-based data gathering. The data gathered from the CH by the sink node are employed dynamically using the mobile sink-based data collection. For this, the MSPP is introduced for gathering the information from the CH using a novel optimization technique. The proposed system model is depicted in Fig. 1.

Initially, the nodes in the WSN are deployed using the Voronoi to solve the coverage issue. Then, the energy-efficient LEACH-based clustering is devised for clustering and CH selection for energy-efficient routing. Then, the MSPP is employed using the extended Aquila (ExAq) optimization algorithm. The multi-objective fitness factors such as delay,

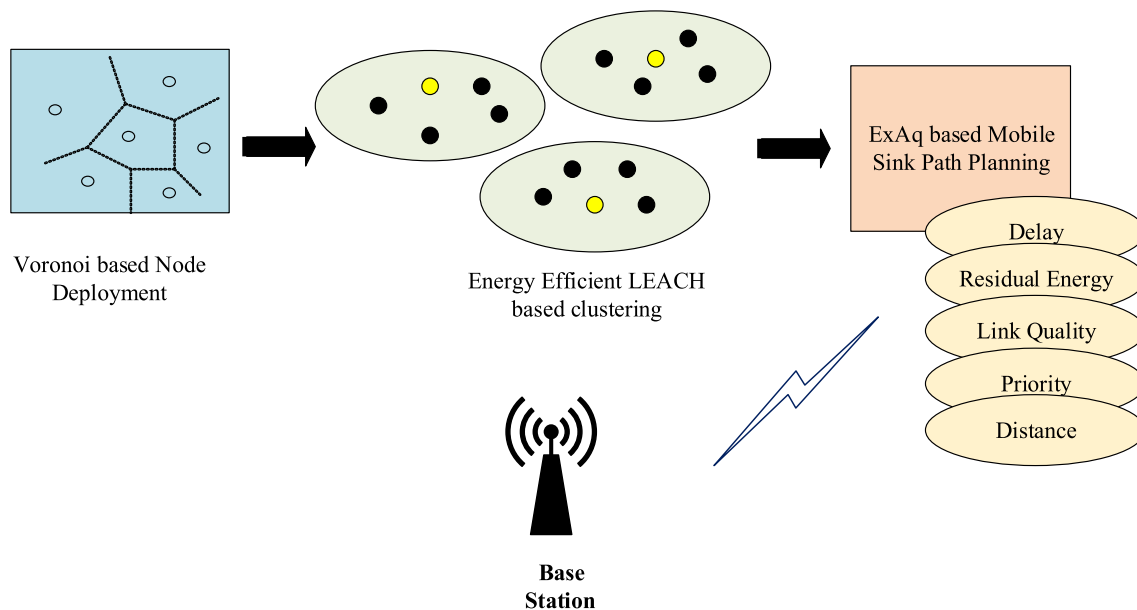


Fig. 1 Proposed ExAq-MSPP-based routing system model

residual energy, link quality, priority, and distance are considered for identifying the optimal best path.

3.1 Node Deployment Using Voronoi

Voronoi-based deployment ensures that sensor nodes evenly cover the entire deployment area. This helps minimize coverage gaps and provides comprehensive data collection. In this, nodes are strategically placed, reducing the energy required for long-distance communication. This can significantly extend the network's lifetime as sensor nodes conserve energy. Besides, by assigning specific regions to nodes, interference and contention for the communication channel are minimized. This results in more efficient data transmission and reduced packet collisions. Thus, in the proposed routing protocol, Voronoi-based node deployment is introduced. The Voronoi-based node deployment is depicted in Fig. 2.

The steps utilized for the Voronoi-based node deployment are:

- Utilize the Delaunay triangulation algorithm to calculate the Delaunay triangulation of the selected node locations. Delaunay triangulation connects nodes in a way that minimizes the occurrence of obtuse angles in triangles.
- Transform the Delaunay triangulation into a Voronoi diagram. You can achieve this by connecting the circumcenters of adjacent triangles in the Delaunay triangulation, creating the edges of the Voronoi diagram.
- For each sensor node, determine which Voronoi region it belongs to. Each node should be associated with the Voronoi region containing its initial location. These Voronoi regions represent the areas of responsibility for the nodes.

Thus, the Voronoi-based node deployment is employed, and then node clustering and CH selection are devised using the LEACH approach for energy-efficient routing.

3.2 Clustering Using Energy-Efficient LEACH

The LEACH algorithm is a powerful tool that enhances the energy efficiency and overall performance of wireless sensor networks. By dynamically selecting CHs, optimizing energy consumption, and strategically organizing data gathering and sharing, it significantly extends the network's lifetime and ensures efficient operation. Initially, the network is divided into various groups or clusters. Each cluster functions as a self-contained unit within the WSN. Within each cluster, one node is designated as the CH. The CH plays a central role in the cluster's operations. The CH is responsible for gathering data from all the cluster members and sharing this collected information with the central sink node of the network. Unlike static assignments where a CH remains unchanged for extended periods, the LEACH algorithm introduces a dynamic CH selection process. During every communication round, a new CH is selected. This dynamic selection enhances the network's lifetime. It is important to note that the node chosen as the CH consumes more energy compared to the other cluster members. This higher energy usage is due to the CH's active involvement in data gathering and sharing with the sink node. The cluster members, on the other hand, have a different role. Their primary responsibility is to share data with the CH. They do not engage in the extensive data gathering and sharing activities performed by the CH. The process of cluster formation and CH selection is initially dynamic and arbitrary, ensuring flexibility and adaptability. After completing the first communication round, the LEACH algorithm employs a probability-based approach to select the CH for the next round. This selection takes into account the energy levels of the nodes, ensuring a balanced distribution of energy usage across the network.

Thus, for the energy-efficient LEACH-based clustering U represents the number of clusters, V signifies the total number of nodes within the network, and Z denotes the probability of selecting a CH. When communication is initiated, each node in the cluster generates a random number between 0 and 1. This number is subsequently compared

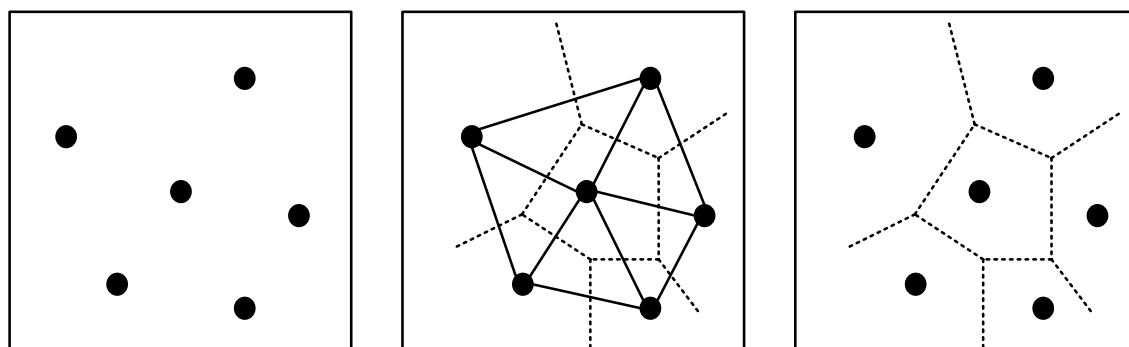


Fig. 2 Voronoi-based node deployment

with a pre-established threshold. The formula for computing this threshold is as follows:

$$T(V) = \begin{cases} \frac{Z}{1-Z(p \bmod \frac{1}{Z})} & \text{for all } V \text{ in cluster} \\ 0, & \text{Else} \end{cases} \quad (1)$$

where p stands for the selection round and $T(V)$ stands for the threshold.

Hence, in the network, all nodes have an equal opportunity to become the CH, regardless of their energy levels. This approach can pose challenges, as nodes with minimal energy may become CHs and quickly deplete their energy, limiting the network's lifetime. To address this issue, the enhanced LEACH algorithm is incorporated into the proposed optimal mobile sink placement protocol, aiming to extend the network's lifetime. Consequently, in the second communication round, the threshold value is recalibrated, taking into account the nodes' residual energy, and is represented as

$$T(V) = \begin{cases} \frac{Z}{1-Z(p \bmod \frac{1}{Z})} \times \frac{X_{Ei}}{X_{Ei}} D, & \text{for all } V \text{ in cluster} \\ 0, & \text{Else} \end{cases} \quad (2)$$

where X_{Er} stands for a node's residual energy, X_{Ei} stands for energy at the beginning of the process, and D stands for the ideal cluster number. The following criteria are used to evaluate the selection of the ideal cluster:

$$D = \sqrt{\frac{V}{2\pi}} \sqrt{\frac{X_{Efs}}{X_{Eamp} e^4 (2U - 1) X_{Ei} - U \cdot X_{Eagg}}} \times P \quad (3)$$

where P refers to the diameter of the network, and the energy expended by the amplifier is indicated as X_{Eamp} . The energy associated with free space communication is represented as X_{Efs} , and the energy consumed by the cluster head (CH) for aggregating information shared by cluster members is denoted as X_{Eagg} .

Using the threshold value estimation, CHs are chosen to facilitate energy-efficient information sharing. In addition, CHs exclusively communicate with the sink node, reducing communication links between cluster members and the sink. This approach ensures minimal communication links, and consequently, the collected data are relayed to the sink through the CHs, making them responsible for sharing information with the mobile sink.

3.3 Extended Aquila Optimization Algorithm-Based MSPP

After clustering the nodes in the network and CH selection, the information gathered by the CH from the cluster nodes is shared with the sink. The consideration of mobile sink is

compared to the static sink due to the benefits of enhanced energy efficiency and network lifetime. Static sinks may require sensor nodes to transmit data over longer distances, leading to higher energy consumption for data transmission. Mobile sinks can move closer to the CH, reducing the energy required for data transmission, which in turn enhances the network lifetime. In the proposed routing protocol, the ExAq optimization algorithm is considered for mobile sink path planning. The ExAq is designed by hybridizing the conventional Aquila optimization with the chaotic Chebyshev mapping for enhancing the exploration criteria with a fast convergence rate. The multi-objective fitness function is considered for identifying the optimal best path for the mobile sink to gather information from the CH. Factors such as delay, residual energy, link quality, priority and distance are utilized for the multi-objective fitness evaluation.

3.3.1 Multi-objective Fitness Function

The multi-objective fitness considered for the MSPP using the ExAq algorithm is evaluated based on various factors such as delay, residual energy, link quality, priority and distance.

Delay: The time taken by the CH to share information with the mobile sink is defined as delay, which is evaluated based on transmission delay T_D , queuing delay Q_D and processing delay P_D . It is formulated as

$$F_D = T_D + P_D + Q_D \quad (4)$$

where the delay associated with the information sharing is defined as F_D .

Residual energy: The available energy resources that can be utilized for communication and data transmission to the mobile sink are measured through the residual energy estimation. It is outlined as

$$F_{RE} = E_{Initial} - E_{Consumed} \quad (5)$$

Table 2 Simulation parameters

Parameters	Values
Nodes	50–150
Data rate	1 Kbps
Transmission range	200 m
Area	1000 × 1000 m
Initial energy	1 J
Traffic	CBR
Population	20
Iteration	1000
Rounds	2500
Node distribution	Randomly distributed
Cluster head	4
Initial pause time	0.5 s

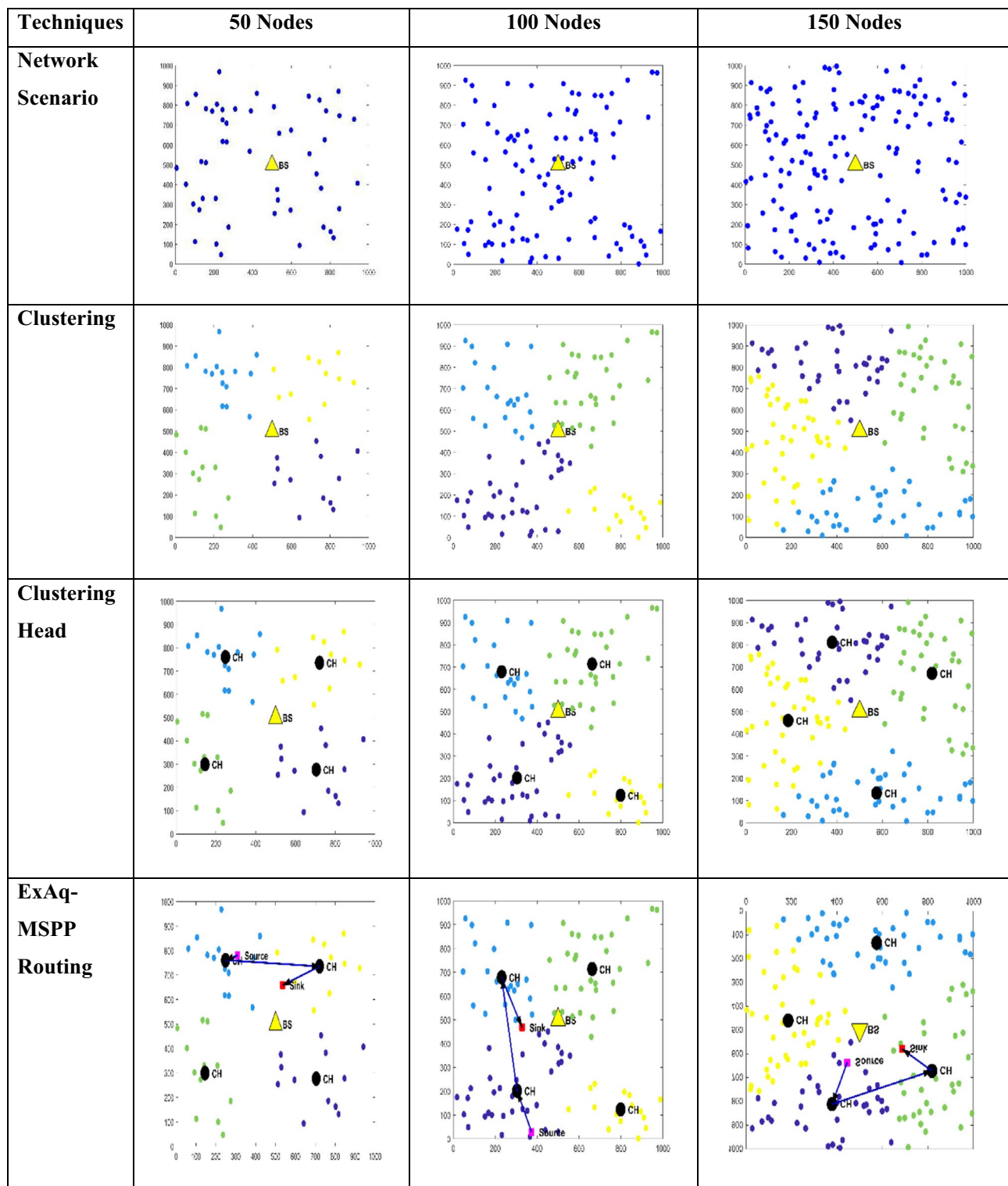


Fig. 3 Simulation results

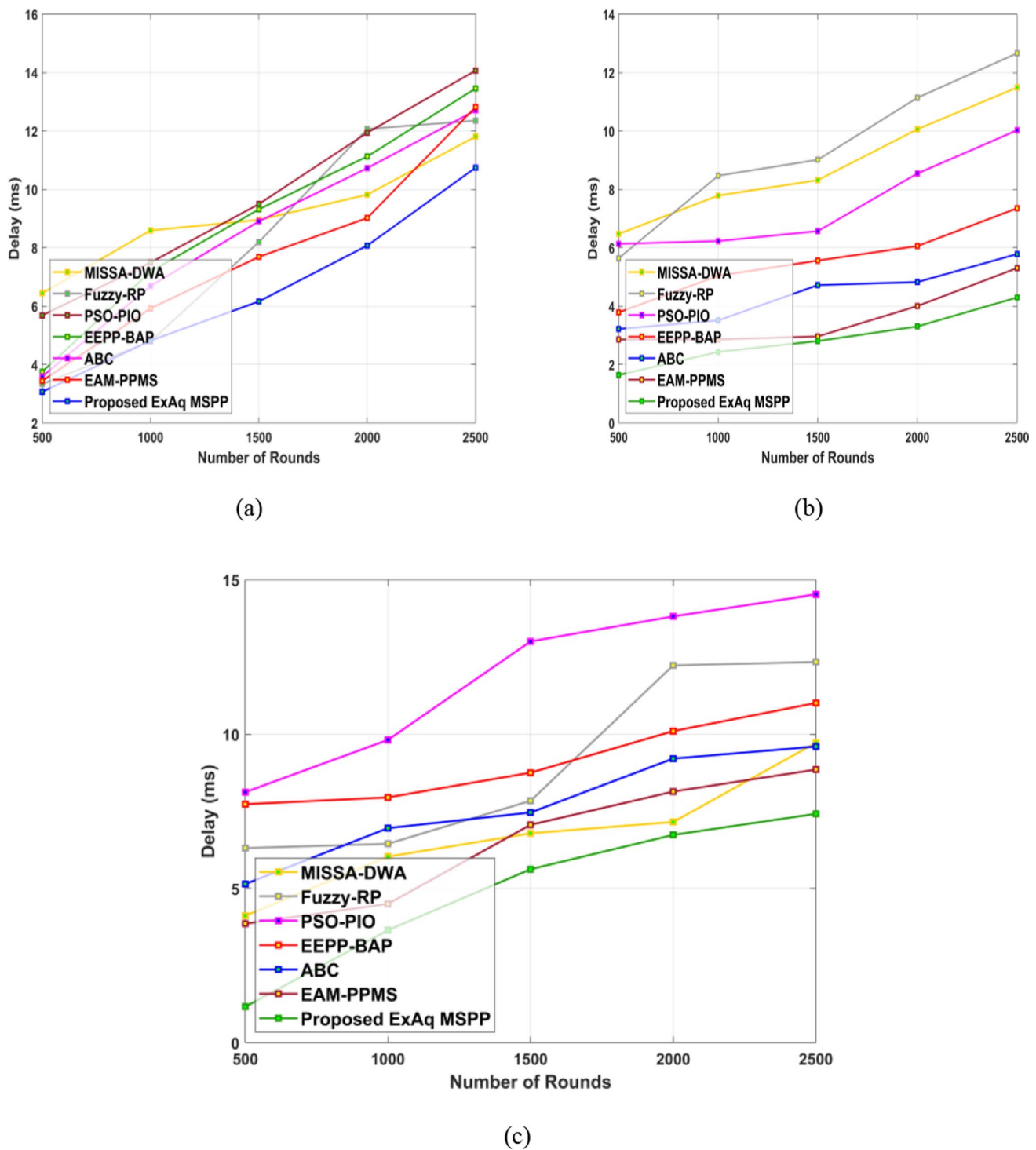


Fig. 4 Delay-based assessment: **a** 50 nodes, **b** 100 nodes and **c** 150 nodes

where the residual energy is defined as F_{RE} , the initial energy is defined as $E_{Initial}$ and the energy consumed is defined as $E_{Consumed}$.

Link quality: Link quality estimation in the mobile sink path planning for data gathering from CHs in a WSN refers

to the process of assessing the reliability and effectiveness of the communication links between CHs and the mobile sink. It helps in determining the quality of the paths through which data are collected from CHs and transmitted to the mobile sink:

$$F_{LQ} = \frac{RSSI}{\text{Noise}} \quad (6)$$

where the link quality is defined as F_{LQ} , and the received signal strength is denoted as RSSI.

Priority: Priority refers to the process of assigning priorities to different CHs or data sources based on various criteria. These priorities help in determining the order or importance of collecting data from CHs and optimizing data transmission in the network. It is denoted as F_p .

Distance: Distance estimation is a critical factor in path planning because it helps determine the optimal route for the mobile sink to traverse the network and collect data efficiently. Distance refers to the process of determining the physical distance between the mobile sink and the CHs. It is formulated as

$$F_{\text{Dist}} = \sqrt{(x_{\text{sink}} - x_{\text{CH}})^2 + (y_{\text{sink}} - y_{\text{CH}})^2} \quad (7)$$

where the distance measure is denoted, the position of the sink is defined as x_{sink} , y_{sink} , the position of the CH is denoted as x_{CH} , y_{CH} and the coordinates are defined as (x, y) .

The formulation for the multi-objective fitness function is outlined as follows:

$$\text{Fit} = \text{Min}\{F_D, F_{\text{Dist}}\}; \text{Max}\{F_{\text{RE}}, F_{LQ}, F_p\} \quad (8)$$

3.3.2 ExAq Algorithm

The proposed ExAq algorithm is designed by integrating the chaotic Chebyshev mapping within the conventional Aquila optimization for enhancing the convergence rate by solving the issue of local optimal solution trapping. The inclusion of chaotic mapping assists in enhancing the exploration, and hence, the global best is accomplished.

Initialization: The population of the solution H is first established together with the issue's lower bound A and upper bound G . The following is an expression for the solutions represented in the search area:

$$H = \begin{bmatrix} h_{1,1} & \cdots & h_{1,l} & h_{1,L-1} & h_{1,L} \\ h_{2,1} & \cdots & h_{2,l} & \cdots & h_{2,L} \\ \cdots & \cdots & h_{k,l} & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ h_{B-1,1} & \cdots & h_{B-1,l} & \cdots & h_{B-1,L} \\ h_{B,1} & \cdots & h_{B,n} & h_{B,L-1} & h_{B,L} \end{bmatrix} \quad (9)$$

The k^{th} location of the Aquila is denoted as h_k , the dimension of the search area is denoted as L , and B stands for the total number of solutions. Following that, the expression that is used to represent the Aquila in the search area:

$$H_{k,l} = M \times (G_l - A_l) + A_l, k = 1, 2, \dots, B; l = 1, 2, \dots, L \quad (10)$$

The upper and lower bounds are denoted as G_l and A_l , respectively, and the random number is denoted as M .

Fitness estimation: The effectiveness or closeness of the solution for the MSPP routing protocol is evaluated using a fitness estimation method. This fitness estimation is quantified using the multi-objective fitness using Eq. (8).

Four hunting tactics: Aquila employs four distinct hunting tactics, encompassing both exploration and exploitation phases, to capture prey. The solution obtained through a balanced approach yields the global best solution. In the diversification phase, attributes within the sight area are explored, resulting in the representation of the global best solution. Subsequently, the intensification phase involves a deeper exploitation of attributes within the sight area, leading to the discovery of the local best solution. Consequently, a solution achieved through a balanced combination of exploration and exploitation aids in resolving the issue related to MSPP by attaining the global best solution.

- (i) **Improved exploration:** During this phase, Aquila intensively explores the sight area for prey capture, utilizing a vertical stoop maneuver initiated from a high soaring altitude. The mathematical expression of Aquila's position, enhanced by randomization, is as follows:

$$H_1(Q+1) = H_{\text{better}}(Q) \times \left(1 - \frac{Q}{Q_{\text{max}}}\right) + (H_F(H) - H_{\text{better}}(Q) * M) \quad (11)$$

where Aquila's location in the following iteration $(Q+1)$ is designated as $H_1(Q+1)$ and the best solution from the current iteration is denoted by $H_{\text{better}}(Q)$. Aquila uses a control element called $\left(1 - \frac{Q}{Q_{\text{max}}}\right)$ to investigate the surroundings in front of it. The result of the current iteration's solution acquisition is summed up, and the average is denoted by the letter $H_F(Q)$ and a random number between 0 and 1 is designated as M . The formulation for the average solution is outlined as follows:

$$H_F(Q) = \frac{1}{B} \sum_{k=1}^B H_k(Q), \forall l = 1, 2, \dots, L \quad (12)$$

- (ii) **Targeted exploration:** During this phase, Aquila conducts circular movements around the prey, transitioning from a high soaring position to a gliding contour flight approach. This phase is appropriately labeled as the preparatory stage for prey capture, and the position of Aquila during this stage is mathematically expressed as

$$H_2(Q) = H_{\text{better}}(Q) \times \text{Levy}(D) + H_S(Q) + (c - d) * M \quad (13)$$

The best Aquila location in the current iteration is marked as $H_{\text{better}}(Q)$. The Aquila's position in the narrower diversification is denoted as $H_2(Q)$ and the spiral-shaped search is denoted by c and d , respectively. The Aquila's Levy flight is denoted as $\text{Levy}(D)$. In addition, the Aquila's arbitrary location between $[1, B]$ is denoted as $H_S(Q)$. Using these notations, we create the following expression for the Levy flight:

$$\text{Levy}(D) = f \times \frac{r \times \alpha}{|s|^{1/\beta}} \quad (14)$$

If r and s are random numbers, and $[0,1]$ is their range, and 0.01 is the value set for the constant f . β is then stated as

$$\alpha = \left(\frac{\Gamma(1 + \beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \right) \quad (15)$$

Table 3 Delay-based assessment

Rounds/methods	500	1000	1500	2000	2500
50 nodes					
MISSA-DWA	6.444	8.594	8.956	9.818	11.812
Fuzzy-RP	3.332	4.794	8.196	12.070	12.355
PSO-PIO	5.684	7.494	9.496	11.941	14.068
EEPP-BAP	3.752	7.181	9.317	11.128	13.456
ABC	3.590	6.691	8.904	10.727	12.707
EAM-PPMS	3.440	5.925	7.687	9.022	12.821
Proposed ExAq MSPP	3.066	4.819	6.162	8.074	10.742
100 nodes					
MISSA-DWA	6.471	7.787	8.315	10.061	11.488
Fuzzy-RP	5.637	8.468	9.013	11.137	12.663
PSO-PIO	6.133	6.228	6.570	8.543	10.025
EEPP-BAP	3.785	5.041	5.559	6.058	7.353
ABC	3.217	3.510	4.720	4.825	5.781
EAM-PPMS	2.855	2.858	2.962	4.004	5.303
Proposed ExAq MSPP	1.641	2.430	2.803	3.304	4.297
150 nodes					
MISSA-DWA	4.117	6.032	6.788	7.157	9.723
Fuzzy-RP	6.310	6.448	7.843	12.232	12.344
PSO-PIO	8.122	9.818	13.009	13.819	14.529
EEPP-BAP	7.733	7.954	8.753	10.103	11.013
ABC	5.145	6.956	7.465	9.211	9.603
EAM-PPMS	3.860	4.501	7.064	8.143	8.853
Proposed ExAq MSPP	1.169	3.649	5.625	6.737	7.421

In this case, β has a fixed value of 1.5. The criteria used to indicate the spiral search are then represented as follows:

$$c = x \times \cos(\lambda) \quad (16)$$

$$d = x \times \sin(\lambda) \quad (17)$$

where

$$x = x_1 + W \times D_1 \quad (18)$$

$$\lambda = -\gamma \times D_1 + \lambda_1 \quad (19)$$

$$\lambda_1 = \frac{3\pi}{2} \quad (20)$$

The fixed values are assigned for some of the constants such as $\gamma = 0.005$, and $W = 0.00565$. The value of D_1 ranges between $[1, B]$ and x_1 refers to the search cycle and ranges between $[1, 20]$.

- (iii) *Combined exploitation*: In this stage, Aquila gets ready to attack the prey using the slow descent and low flight criteria. The mathematical expression for the solution in this phase is expressed as

$$H_3(Q + 1) = (H_{\text{better}}(Q) - H_F(Q)) \times \omega - M + ((G - A) \times M + A) \times \delta \quad (21)$$

where the parameters used to control the intensity are denoted as ω and δ , each having a value of 0.1. To enhance the exploration criteria, the chaotic Chebyshev mapping is devised and formulated as follows:

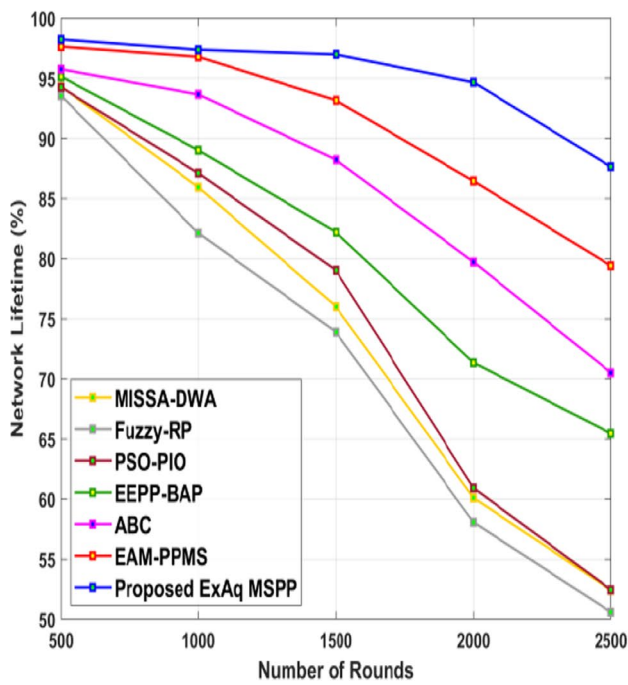
$$H_{Q+1} = \cos(n \cdot \cos^{-1} H_Q) \quad (22)$$

where H_{Q+1} denotes the solution reached in the current iteration and H_Q denotes the solution reached in the previous iteration Q . The control parameter used to choose the most suitable candidates is n , and it has a value between 0 and 1. Thus, the position updation in this phase using the ExAq algorithm is expressed as

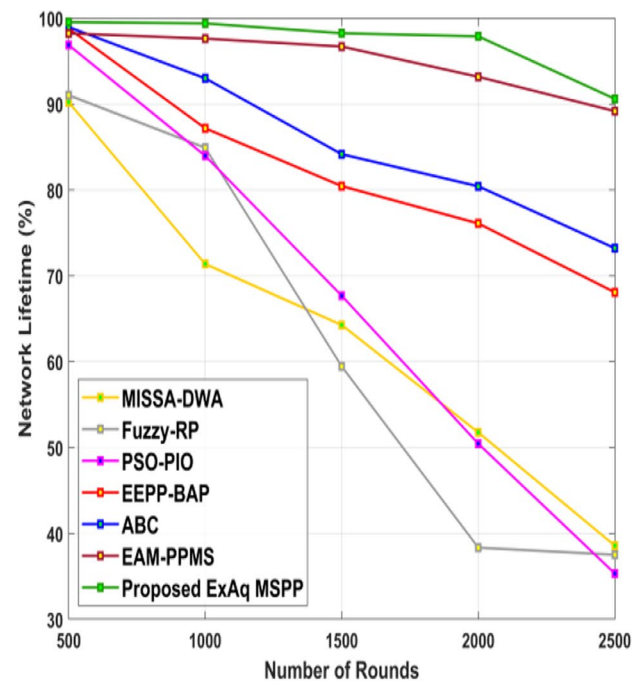
$$W_3(T + 1) = 0.5 \{ (H_{\text{better}}(Q) - H_F(Q)) \times \omega - M + ((G - A) \times M + A) \times \delta \} + 0.5 \{ \cos(n \cdot \cos^{-1} H_Q) \} \quad (23)$$

Thus, using the proposed solution updation helps to acquire the local best solution from the global best solution to obtain the balanced phases.

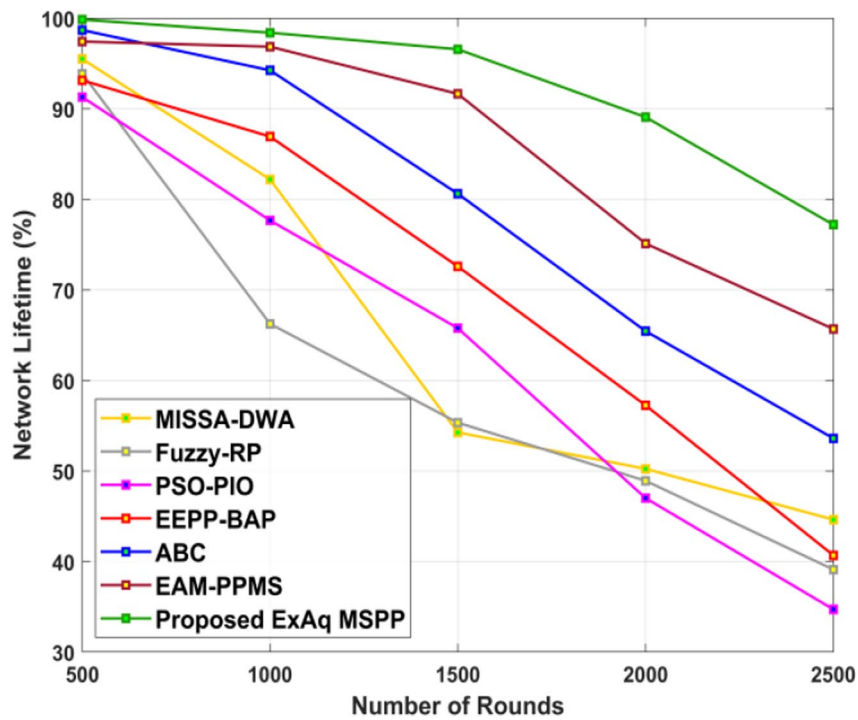
- (iv) *Acquisition of target*: In this stage, Aquila captures the prey by walking, and the expression for the solution is represented as



(a)



(b)



(c)

Fig. 5 Network lifetime-based assessment: **a** 50 nodes, **b** 100 nodes and **c** 150 nodes

$$H_4(Q+1) = J \times H_{\text{better}}(Q) - (T_1 \times H(q) \times M) - T_2 \times \text{Levy}(D) + M \times T_1 \quad (24)$$

where the quality function, Aquila's movement to catch its prey, and the slope of flight are each denoted by T_2 , T_1 , and J , and the formula for determining these parameters is written as

$$J(Q) = Q^{\left(\frac{2 \times M(Q)-1}{(1-Q)^2}\right)} \quad (25)$$

$$T_1 = 2 \times M \left(\frac{Q}{Q_{\max}} \right) - 1 \quad (26)$$

$$T_2 = 2 \times \left(1 - \frac{Q}{Q_{\max}} \right) \quad (27)$$

If the value of the random number M spans from $[0,1]$, the current iteration is denoted by Q , and the maximum iteration is denoted by $Q_{\max}$.

Re-evaluating the fitness: Following the adjustment of Aquila's positions, the fitness is re-assessed to verify the solution's feasibility of MSPP, employing Eq. (8).

Termination: After the accomplishment of $Q_{\max}$ or the acquisition of the global best solution, the iteration terminates and the solution is utilized for routing the request. The pseudo-code is presented in Algorithm 1.

Algorithm 1 Pseudocode for ExAq algorithm

Pseudocode for ExAq Optimization	
1	Initialize the maximum iteration $Q_{\max}$, population B and, parameters α , W etc.
2	{
3	While ($(Q_{\max} > Q)$) do
4	Calculate the fitness
5	Determine the best solution H_{better}
6	for ($k = 1, 2, \dots, B$) do
7	Estimate the $H_F(Q)$
8	Using equation (11), update the solution.
9	Using equation (13), update the solution.
10	Using the novel ExAq equation (23) based solution updation
11	Using equation (24), update the solution.
12	Re-estimate the fitness
13	Return the best solution.
14	$Q = Q + 1$
15	}
16	end

Table 4 Network lifetime-based assessment

Rounds/methods	500	1000	1500	2000	2500
50 nodes					
MISSA-DWA	94.321	85.946	76.013	60.105	52.421
Fuzzy-RP	93.545	82.127	73.890	58.070	50.595
PSO-PIO	94.259	87.109	79.025	60.924	52.455
EEPP-BAP	95.146	89.020	82.202	71.348	65.460
ABC	95.754	93.659	88.226	79.724	70.512
EAM-PPMS	97.640	96.795	93.164	86.452	79.404
Proposed ExAq-MSPP	98.240	97.379	96.991	94.672	87.640
100 nodes					
MISSA-DWA	90.211	71.380	64.270	51.769	38.577
Fuzzy-RP	91.029	84.913	59.446	38.345	37.521
PSO-PIO	96.875	83.972	67.679	50.448	35.337
EEPP-BAP	98.792	87.182	80.449	76.101	68.079
ABC	98.966	93.000	84.160	80.428	73.221
EAM-PPMS	98.233	97.628	96.699	93.175	89.179
Proposed ExAq-MSPP	99.544	99.396	98.251	97.888	90.609
150 nodes					
MISSA-DWA	95.506	82.208	54.261	50.240	44.632
Fuzzy-RP	93.876	66.252	55.337	48.918	39.118
PSO-PIO	91.301	77.675	65.773	47.006	34.720
EEPP-BAP	93.152	86.937	72.592	57.252	40.687
ABC	98.694	94.256	80.623	65.446	53.606
EAM-PPMS	97.435	96.865	91.658	75.127	65.683
Proposed ExAq-MSPP	99.857	98.408	96.580	89.085	77.207

Thus, using the solution accomplished using the ExAq algorithm, MSPP is employed optimally for enhancing the network lifetime through the multi-objective fitness evaluation and chaotic Chebyshev mapping.

4 Results and Discussion

The proposed MSPP approach based on ExAq is implemented in MATLAB and is assessed based on various assessment measures such as delay, network lifetime, packet delivery ratio, residual energy and throughput. To indicate the superiority of the proposed ExAq-MSPP technique, the conventional MSPP-based routing protocols such as PSO-PIO [1, 2] EEPP-BAP [29] ABC [30], EAM-PPMS [31], MISSA-DWA [33], and Fuzzy-RP [34]. The simulation parameters of the proposed ExAq-MSPP technique are depicted in Table 2.

4.1 Simulation Results

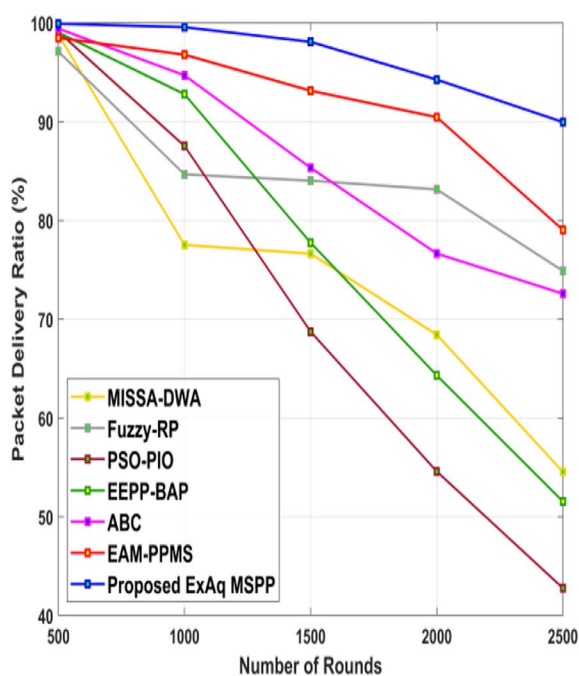
Figure 3 serves as a detailed visual representation of the ExAq-MSPP algorithm's experimental outcomes, illustrating its performance across varying network configurations with node densities of 50, 100, and 150. Figure 3 encapsulates the core processes of the algorithm, including network scenarios, clustering procedures, cluster head selection, and routing strategies. The network scenarios depicted in the figure provide a comparative overview of how the algorithm operates under different conditions, showcasing its adaptability to various topologies and node distributions. The clustering process is intensely represented, with each cluster clearly defined, allowing us to observe how nodes are organized within the network. Special attention is given to the selection of cluster heads, which are highlighted with distinct symbols to underscore their importance in managing data aggregation and communication within each cluster. Figure 3 also defines the routing paths, using arrows to visualize the flow of data from cluster heads to the mobile sink, emphasizing the algorithm's efficiency in minimizing energy consumption and latency during transmission. By presenting the simulation results for different node densities side by side, Fig. 3 provides a clear comparison of the algorithm's scalability and effectiveness as the network size increases. This visual summary not only demonstrates the ExAq-MSPP algorithm's robustness but also its potential application in large-scale, energy-constrained environments such as wireless sensor networks, making it an essential component of the research findings.

4.2 Comparative Assessment

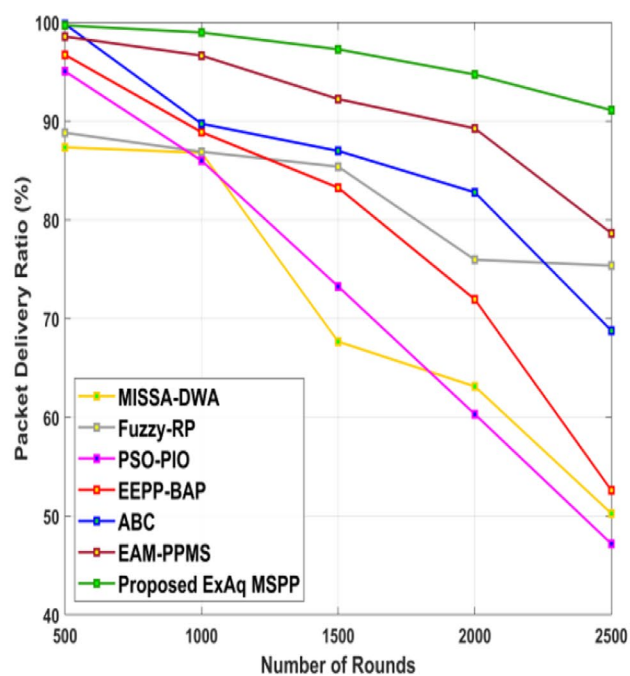
The proposed ExAq-MSPP routing protocol is assessed based on various measures such as delay, network lifetime, packet delivery ratio, residual energy and throughput for various nodes are detailed in this section.

4.2.1 Delay-Based Assessment

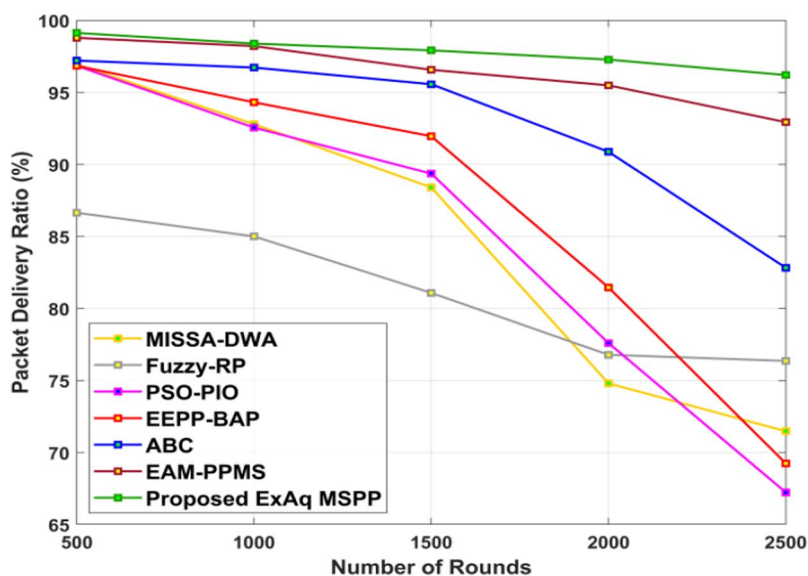
Delay in MSPP for a wireless sensor network refers to the time it takes for a mobile sink to traverse a particular path while collecting data from CH in the network. This delay is a crucial metric in such networks, as minimizing it helps ensure timely data retrieval and efficient network operation.



(a)



(b)



(c)

Fig. 6 Packet delivery ratio-based assessment: **a** 50 nodes, **b** 100 nodes and **c** 150 nodes

Table 5 Packet delivery ratio-based assessment

Rounds/methods	500	1000	1500	2000	2500
50 nodes					
MISSA-DWA	98.814	77.529	76.637	68.429	54.543
Fuzzy-RP	97.114	84.665	84.032	83.144	74.891
PSO-PIO	99.112	87.539	68.739	54.583	42.765
EEPP-BAP	99.032	92.783	77.746	64.308	51.548
ABC	99.429	94.697	85.332	76.649	72.584
EAM-PPMS	98.490	96.787	93.137	90.459	79.046
Proposed ExAq-MSPP	99.920	99.572	98.089	94.258	89.964
100 nodes					
MISSA-DWA	87.350	86.794	67.669	63.133	50.269
Fuzzy-RP	88.820	86.897	85.388	75.965	75.367
PSO-PIO	95.049	85.993	73.257	60.323	47.191
EEPP-BAP	96.717	88.882	83.261	71.949	52.607
ABC	99.828	89.736	86.983	82.791	68.782
EAM-PPMS	98.562	96.636	92.238	89.268	78.645
Proposed ExAq-MSPP	99.697	98.979	97.270	94.727	91.113
150 nodes					
MISSA-DWA	96.950	92.796	88.404	74.799	71.479
Fuzzy-RP	86.643	85.007	81.083	76.782	76.357
PSO-PIO	96.852	92.559	89.365	77.596	67.232
EEPP-BAP	96.849	94.314	91.960	81.461	69.245
ABC	97.212	96.727	95.571	90.878	82.825
EAM-PPMS	98.792	98.216	96.574	95.492	92.939
Proposed ExAq-MSPP	99.134	98.390	97.920	97.279	96.210

The delay-based assessment is depicted in Fig. 4, and its detailed analysis is presented in Table 3.

4.2.2 Network Lifetime-Based Assessment

Network lifetime in MSPP for information gathering from CHs in a WSN refers to the duration for which the network can operate efficiently before the depletion of its energy resources. This is a critical metric, as extending the network lifetime is often a primary goal in such systems. The network lifetime-based analysis for various nodes is depicted in Fig. 5, and the detailed analysis is presented in Table 4.

4.2.3 Packet Delivery Ratio-Based Assessment

Packet delivery ratio (PDR) in MSPP for information gathering from CHs refers to the ratio of successfully delivered packets to the total number of packets

transmitted from the CHs to the mobile sink. PDR is a crucial metric to assess the reliability and efficiency of data transmission in the network. A higher PDR indicates a more reliable network with fewer packet losses during data collection. The PDR-based assessment is depicted in Fig. 6, and a detailed analysis is presented in Table 5.

4.2.4 Residual Energy-Based Assessment

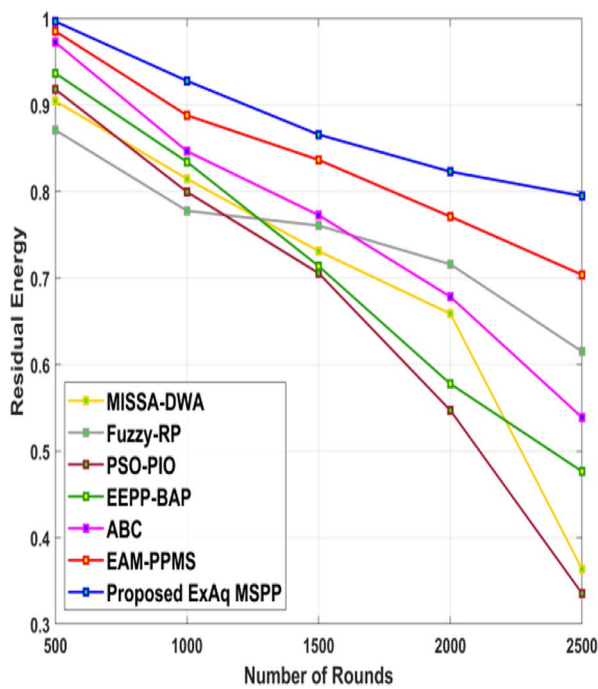
Residual energy in MSPP for information gathering from CHs refers to the amount of energy that remains in the network's sensor nodes after a certain period of operation. This metric is important for assessing the energy status of the network, which can impact the network's sustainability and performance. It helps in making decisions related to routing, data collection, and mobile sink path planning to prolong the network's lifetime and ensure efficient information gathering from CHs in the wireless sensor network. Figure 7 portrays the residual energy-based analysis and its detail is presented in Table 6.

4.2.5 Throughput-Based Assessment

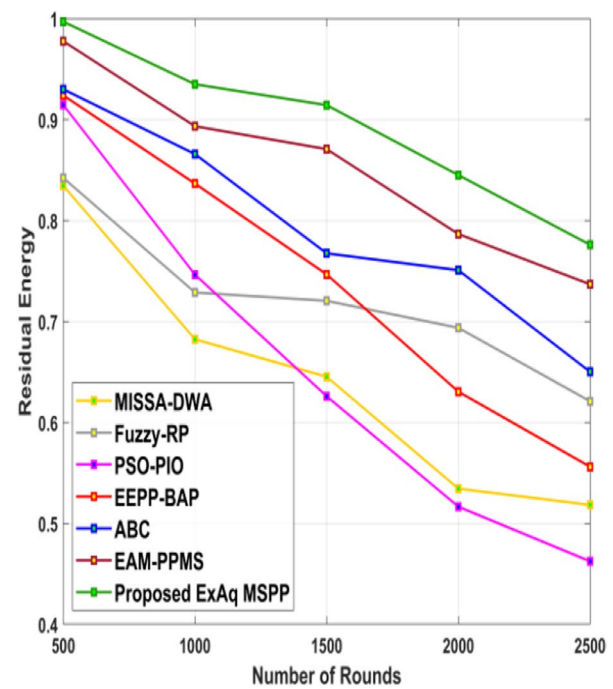
Throughput in MSPP for information gathering from CHs refers to the rate at which data are successfully transmitted from the CHs to the mobile sink over the network. It measures the efficiency of data transfer and reflects the network's capacity for handling data. Throughput is a critical performance metric because it indicates how efficiently the network can transfer data to the mobile sink while considering factors such as network congestion, packet losses, and the speed of the mobile sink. A higher throughput implies that data is being transmitted more quickly and efficiently from the CHs to the mobile sink, which can be essential for timely and reliable information gathering in a wireless sensor network. Throughput-based analysis is illustrated in Fig. 8 and its detailed analysis is presented in Table 7.

4.2.6 Convergence Analysis

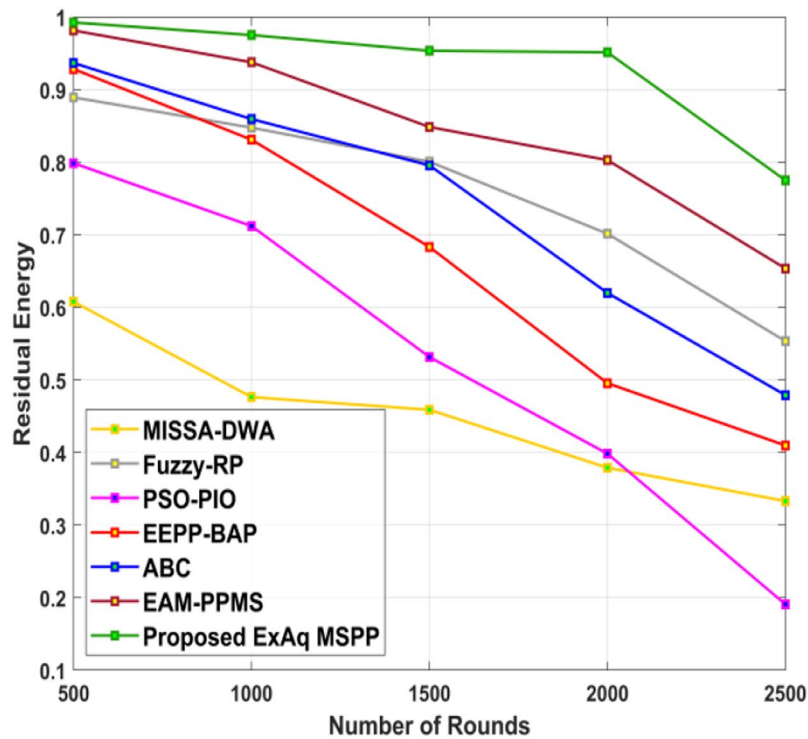
The convergence analysis of the proposed ExAq algorithm and the existing Aquila optimization algorithm is presented in Fig. 9. The analysis portrays that the convergence rate of the proposed ExAq algorithm is faster compared to the existing approach due to the incorporation of chaotic Chebyshev mapping.



(a)



(b)



(c)

Fig. 7 Residual energy-based assessment: **a** 50 nodes, **b** 100 nodes and **c** 150 nodes

Table 6 Residual energy-based assessment

Rounds/methods	500	1000	1500	2000	2500
50 nodes					
MISSA-DWA	0.905	0.815	0.731	0.659	0.364
Fuzzy-RP	0.871	0.778	0.761	0.716	0.615
PSO-PIO	0.918	0.800	0.706	0.547	0.336
EEPP-BAP	0.937	0.834	0.714	0.578	0.477
ABC	0.973	0.847	0.773	0.679	0.539
EAM-PPMS	0.985	0.888	0.837	0.771	0.704
Proposed ExAq-MSPP	0.997	0.928	0.866	0.823	0.795
100 nodes					
MISSA-DWA	0.834	0.682	0.645	0.534	0.518
Fuzzy-RP	0.842	0.729	0.720	0.694	0.621
PSO-PIO	0.915	0.746	0.626	0.517	0.462
EEPP-BAP	0.924	0.837	0.747	0.630	0.556
ABC	0.930	0.866	0.768	0.751	0.650
EAM-PPMS	0.978	0.893	0.871	0.787	0.737
Proposed ExAq-MSPP	0.997	0.935	0.914	0.845	0.776
150 nodes					
MISSA-DWA	0.608	0.476	0.459	0.379	0.333
Fuzzy-RP	0.889	0.848	0.801	0.702	0.553
PSO-PIO	0.799	0.712	0.532	0.398	0.191
EEPP-BAP	0.929	0.831	0.683	0.496	0.410
ABC	0.937	0.860	0.796	0.620	0.479
EAM-PPMS	0.982	0.938	0.849	0.803	0.654
Proposed ExAq-MSPP	0.993	0.975	0.954	0.952	0.775

4.3 Comparative Discussion

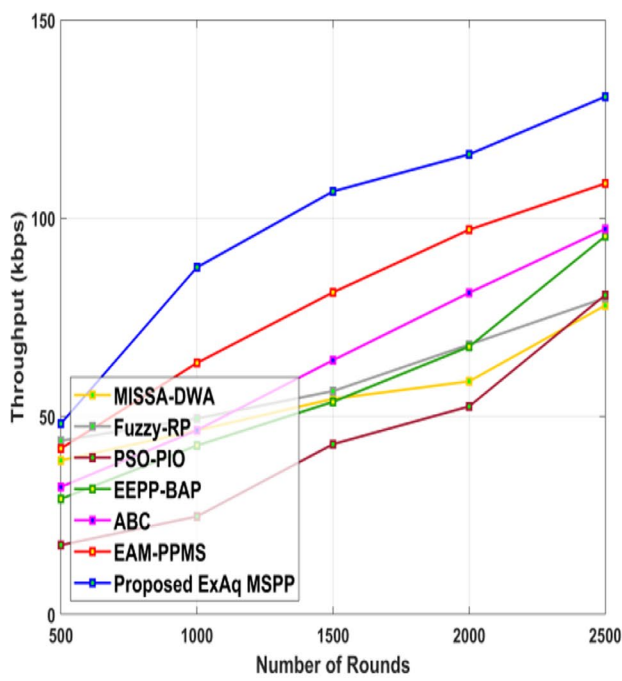
Table 8 depicts the comparative discussion based on the best outcome. The minimal delay evaluated by the ExAq-MSPP is 1.1169, which is 71.61%, 81.48%, 85.61%, 84.88%, 77.28%, and 69.72% elevated outcome compared to the MISSA-DWA, Fuzzy-RP, PSO-PIO, EEPP-BAP, ABC, and EAM-PPMS. The higher network lifetime evaluated by the ExAq-MSPP is 99.857%, which is 4.36%, 5.99%, 8.57%, 6.71%, 1.16%, and 2.43% elevated outcome compared to the MISSA-DWA, Fuzzy-RP, PSO-PIO, EEPP-BAP, ABC, and EAM-PPMS. The higher packet delivery ratio evaluated by the ExAq-MSPP is 99.920%, which is 1.11%, 2.81%, 0.81%, 0.89%, 0.49%, and 1.43% elevated outcome compared to the MISSA-DWA, Fuzzy-RP, PSO-PIO, EEPP-BAP, ABC, and EAM-PPMS. The higher residual energy evaluated by the ExAq-MSPP is 0.997, which is 9.23%, 12.60%, 7.88%, 6.01%, 2.42%, and 1.13% elevated outcome compared to the MISSA-DWA, Fuzzy-RP, PSO-PIO, EEPP-BAP, ABC, and EAM-PPMS. The higher throughput evaluated by the ExAq-MSPP is 255.306, which is 68.12%, 68.20%, 39.02%, 28.22%, 26.11%, and 9.83% elevated outcome compared to the MISSA-DWA, Fuzzy-RP, PSO-PIO, EEPP-BAP, ABC, and EAM-PPMS.

The proposed ExAq-MSPP approach accomplished superior outcomes compared to the existing methods due to the novel routing protocol. The proposed ExAq addresses these challenges faced by the static sink by intelligently moving sinks through the network to optimize data collection. It offers benefits such as improved coverage, energy efficiency, reduced data latency, enhanced scalability, and adaptability to dynamic scenarios. MSPP ensures that data are collected efficiently, leading to more accurate and timely results in various WSN applications. The analysis of the proposed routing protocol with various numbers of nodes such as 50, 100 and 150 are evaluated to depict the scalability of the model. With the increase in the number of nodes, the performance of the proposed ExAq-MSPP is superior compared to existing methods. Thus, the proposed ExAq-MSPP protocol is scalable to various network sizes.

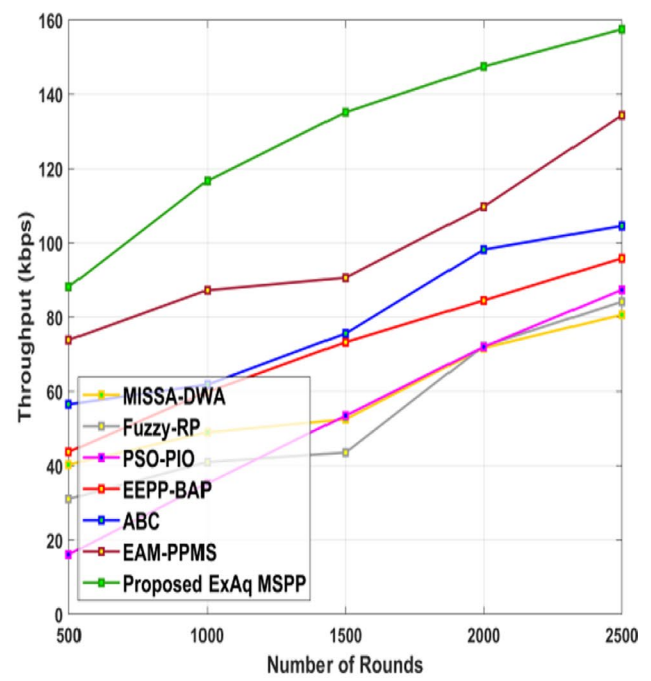
5 Conclusion

A new approach to enhance data gathering in WSNs through the ExAq-MSPP routing protocol is introduced in this research. In this, the Voronoi diagrams to strategically deploy sensor nodes, ensuring uniform coverage, reducing energy consumption, and minimizing interference. The proposed protocol uses the LEACH algorithm for dynamic cluster formation and CH selection, which extends the network's lifetime by optimizing energy usage. The ExAq Optimization algorithm employs a multi-objective fitness function considering delay, residual energy, link quality, priority, and distance that ensures efficient data collection by the mobile sink through the optimal path. Simulation results demonstrate that the ExAq-MSPP protocol outperforms existing protocols in terms of lower delay of 1.169 s, extended network lifetime of 99.857%, higher packet delivery ratio of 99.920%, enhanced residual energy of 0.997 J, and increased throughput of 255.306. This research contributes to the development of efficient routing protocols for WSNs, which are crucial for various applications in monitoring and data collection.

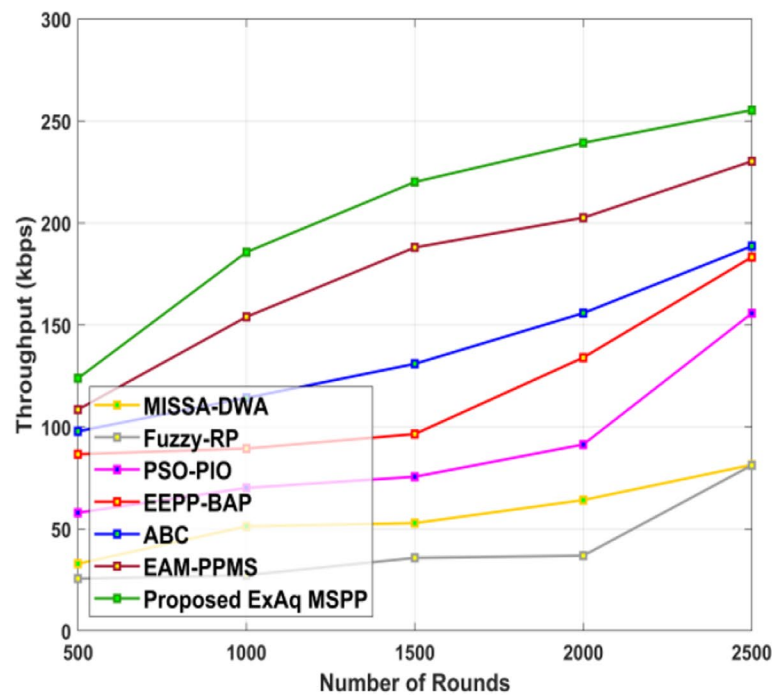
Future research could focus on integrating machine learning techniques into the ExAq-MSPP protocol to dynamically adjust the multi-objective fitness function based on real-time network conditions, further optimizing energy consumption and improving overall network performance. In addition, exploring the application of ExAq-MSPP in heterogeneous WSN environments with varying node capabilities could test the protocol's robustness and adaptability. Another path could involve scaling the protocol for large-scale urban or industrial WSNs,



(a)



(b)



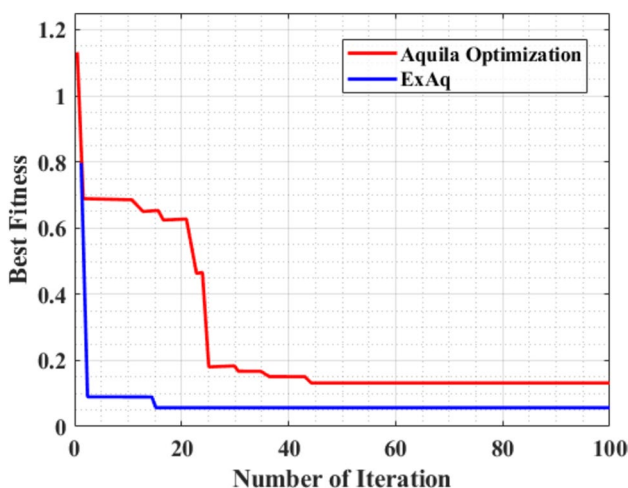
(c)

Fig. 8 Throughput-based assessment: **a** 50 nodes, **b** 100 nodes and **c** 150 nodes

Table 7 Throughput-based assessment

Rounds/methods	500	1000	1500	2000	2500
50 nodes					
MISSA-DWA	38.881	46.422	54.399	58.814	77.970
Fuzzy-RP	43.829	49.431	56.363	68.086	79.857
PSO-PIO	17.490	24.727	42.964	52.497	80.612
EEPP-BAP	29.153	42.633	53.645	67.571	95.435
ABC	32.186	46.467	64.155	81.173	97.283
EAM-PPMS	41.862	63.461	81.281	97.081	108.736
Proposed ExAq-MSPP	48.138	87.590	106.754	116.102	130.652
100 nodes					
MISSA-DWA	40.313	48.973	52.502	71.683	80.633
Fuzzy-RP	31.049	40.962	43.543	72.216	84.134
PSO-PIO	16.109	35.233	53.485	72.001	87.331
EEPP-BAP	43.727	59.858	73.257	84.505	95.833
ABC	56.529	61.844	75.599	98.183	104.579
EAM-PPMS	73.868	87.228	90.610	109.730	134.351
Proposed ExAq-MSPP	88.175	116.684	135.164	147.469	157.512
150 nodes					
MISSA-DWA	32.910	51.263	52.850	64.150	81.400
Fuzzy-RP	25.593	27.289	35.808	36.877	81.179
PSO-PIO	57.940	70.094	75.569	91.402	155.696
EEPP-BAP	86.613	89.403	96.510	133.998	183.256
ABC	97.848	114.208	130.918	155.794	188.655
EAM-PPMS	108.502	153.945	187.984	202.559	230.219
Proposed ExAq-MSPP	123.805	185.675	220.081	239.279	255.306

addressing challenges such as interference and node density, and enhancing security measures to protect against potential cyber-attacks. Finally, extending the protocol to handle complex data types, such as multimedia, could broaden its applicability to advanced fields such as smart

**Fig. 9** Convergence analysis**Table 8** Comparative discussion

Methods/ metrics	Delay	Network lifetime	Packet delivery ratio	Residual energy	Through- put
MISSA-DWA	4.117	95.506	98.814	0.905	81.400
Fuzzy-RP	6.310	93.876	97.114	0.871	81.179
PSO-PIO	8.122	91.301	99.112	0.918	155.696
EEPP-BAP	7.733	93.152	99.032	0.937	183.256
ABC	5.145	98.694	99.429	0.973	188.655
EAM-PPMS	3.860	97.435	98.490	0.985	230.219
Proposed ExAq-MSPP	1.169	99.857	99.920	0.997	255.306

cities and environmental monitoring, making ExAq-MSPP a versatile tool for future wireless sensor networks.

Author Contributions S. Sangeetha: Conceptualization, Methodology, Software, Data curation, Validation, Formal analysis, writing—original draft, Resources, Visualization, Writing—original draft; T. Aruldoss Albert Victoire: Methodology, Software, Validation, Formal analysis, Investigation, Supervision, Writing—review & editing; Manoharan Premkumar: Formal analysis, Software, Data Curation, Project administration, Validation, Visualization, Writing—review & editing; Ravi-chandran Sowmya: Software, Data Curation, Formal analysis, Project administration, Funding, Writing—review & editing.

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Declarations

Conflict of interest The authors declare no competing interests.

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